



PROJECT SYNOPSIS

ON

RoomEase Hostel Management System

Submitted for

Database Management Systems

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ABSTRACT

The RoomEase Hostel Management System is a database-driven application designed to automate and streamline hostel operations efficiently. The system provides a centralized platform for managing student records, room allocation, fee tracking, complaints, and visitor logs. It ensures structured data storage, role-based access control, and improved operational efficiency. By implementing relational database concepts, the system minimizes redundancy, ensures data integrity, and enables efficient data retrieval. This project demonstrates practical implementation of DBMS concepts such as entity-relationship modelling, normalization, and structured query processing in a real-world hostel management scenario.

1. Introduction

Hostel management in educational institutions involves handling large amounts of data related to students, rooms, staff, and services. Traditional manual systems often lead to inefficiencies, data redundancy, and errors.

RoomEase addresses these issues by providing a centralized and automated database system that simplifies hostel operations such as room allocation, fee management, and complaint handling. The system focuses on structured data storage, efficient retrieval, and maintaining consistency using DBMS principles.

2. Problem Statement

Traditional systems lack structured data management, causing redundancy and inefficiency.

A database-driven system is required to manage hostel operations in a reliable and organized way.

3. Objectives of the Project

- Centralized hostel database
- Efficient room allocation and record management
- Maintain data integrity and reduce redundancy
- Manage fees, complaints, and visitors
- Apply DBMS concepts practically

4. Motivation Behind the Project

Manual hostel systems are inefficient and prone to errors. Managing records such as room allocation, complaints, and visitor logs becomes difficult without a structured system. The motivation behind RoomEase is to develop a simplified, efficient, and automated hostel management system that focuses on core functionalities and improves operational efficiency while demonstrating DBMS concepts clearly .

5. Scope of the Project

The scope of RoomEase includes:

- Managing student records and room allocation
- Tracking fee payments and dues
- Handling complaint registration and resolution
- Maintaining visitor records
- Providing secure access through user authentication

The system focuses on essential hostel operations and is suitable for academic demonstration of DBMS concepts.

6. System Description

RoomEase follows a database-centric architecture where all hostel-related data is stored in a structured relational database. The system ensures:

- Centralized data management
- Role-based access (Admin, Staff, Student)
- Efficient querying and retrieval
- Secure authentication and authorization

The system uses SQL-based operations for inserting, updating, deleting, and retrieving data.

7. Module Description

7.1 Student Management Module

Handles student registration, record updates, and room allocation.

7.2 Room Management Module

Manages room availability, allocation, and status updates (Vacant/Occupied).

7.3 Fee Management Module

Tracks fee payments, due dates, and payment status.

7.4 Complaint Management Module

Allows students to file complaints and tracks their resolution status.

7.5 Visitor Management Module

Maintains visitor records including check-in and check-out details.

7.6 User Authentication Module

Ensures secure login and role-based access for admins, staff, and students.

8. Database Design

8.1 Entities and Attributes

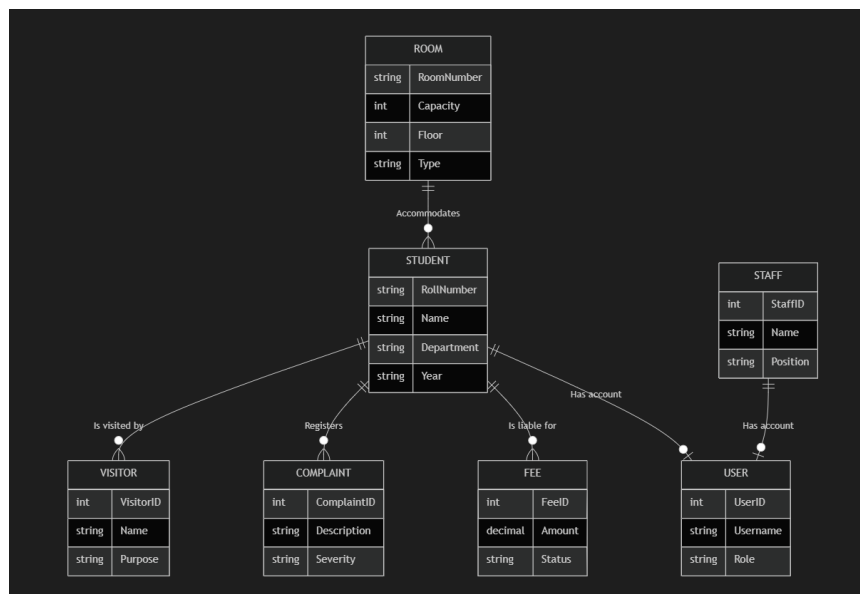
- **Student:** Roll No, Name, Department, Room No, Contact, Email
- **Room:** Room No, Capacity, Status, Floor, Type
- **Staff:** Staff ID, Name, Position, Contact
- **Fees:** Fee ID, Amount, Due Date, Status
- **Complaint:** Complaint ID, Type, Severity, Status
- **Visitor:** Visitor ID, Name, Visit Date, Purpose
- **User:** User ID, Username, Password, Role

8.2 Relationships

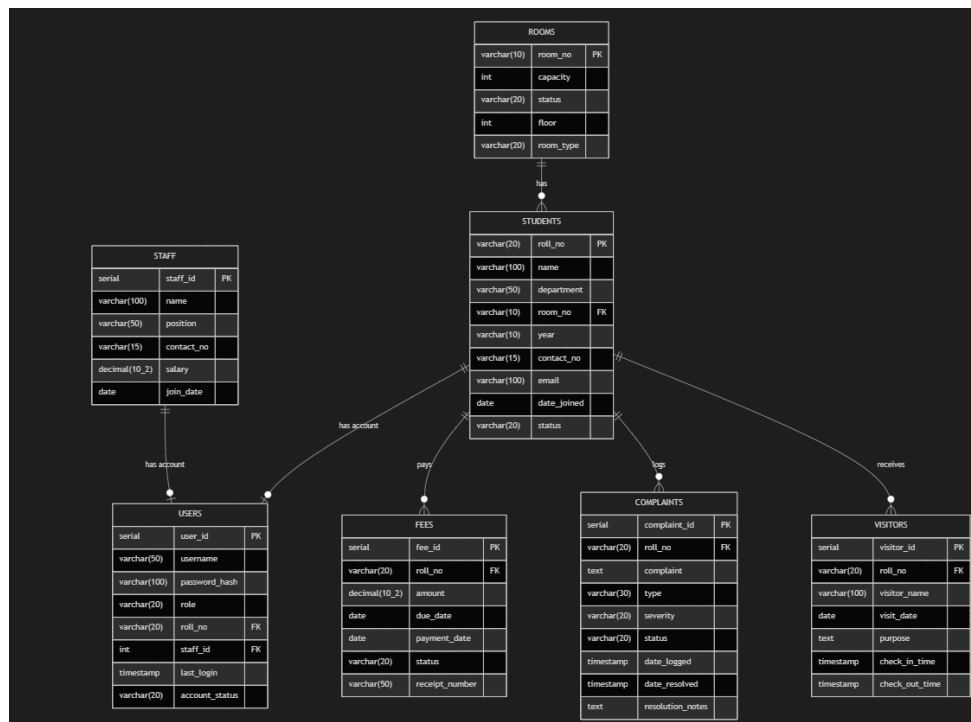
- One student is allocated one room
- One student can file multiple complaints
- One student can have multiple fee records
- One student can receive multiple visitors
- Staff manages hostel services

9. ER Diagram Description

The ER model represents key entities such as Student, Room, Fees, Complaint, Visitor, and Staff, along with their relationship. the Student entity is central and connects with rooms, complaints, fees, and visitors, ensuring structured data organization and efficient querying.



ER model



Relational model

10. Normalisation

The database is normalized up to Third Normal Form (3NF) to:

- Eliminate redundancy
- Ensure data consistency
- Improve efficiency

Each table contains atomic attributes, and all non-key attributes depend fully on the primary key.

11. Technologies Used

Database & Backend Logic: Oracle SQL & PL/SQL:

A relational database management system (RDBMS) used for structured data storage and ensuring data integrity. Server-side logic, data manipulation, and automation are handled entirely inside the database using PL/SQL stored procedures, functions, and triggers.

Tools:

- Oracle Live SQL / SQL Developer – Database environment for executing scripts .
- VS Code – Development environment .
- Git – Version control and collaboration.

12. Conclusion

The RoomEase Hostel Management System successfully demonstrates the application of DBMS concepts in a real-world scenario. It provides a structured, efficient, and reliable system for managing hostel operations. By implementing relational database design, normalization, and automation, the system reduces manual effort, minimizes errors, and improves overall efficiency. The project serves as a strong academic example of database design and management.